

1.2.2: Find the complement c^c of the set c with respect to the space $\mathcal{C}$ if

(a) $\mathcal{C} = \{x: 0 < x < 1\}$, $c = \{x: \frac{5}{8} < x < 1\}$

(b) $\mathcal{C} = \{(x, y, z): x^2 + y^2 + z^2 \leq 1\}$, $c = \{(x, y, z): x^2 + y^2 + z^2 = 1\}$

(c) $\mathcal{C} = \{(x, y): |x| + |y| \leq 2\}$, $c = \{(x, y): x^2 + y^2 < 2\}$

Solution (a)

$$c^c = \{x: 0 < x \leq \frac{5}{8}\}$$

Solution (b)

$$c^c = \{(x, y, z): x^2 + y^2 + z^2 < 1\}$$

Solution (c)

First note that $x^2 + y^2 < 2$ can be written as

$$(|x| + |y|)^2 - 2|x||y| < 2 \quad \text{or} \quad |x| + |y| < \sqrt{2 + 2|x||y|}$$

$$\text{Hence } c^c = \{(x, y): \sqrt{2 + 2|x||y|} \leq |x| + |y| \leq 2\}$$

1.2.6 show that the following sequences of sets $\{C_k\}$ are nondecreasing, (1.2.16), then find $\lim_{k \rightarrow \infty} C_k$

(a) $C_k = \{x: \frac{1}{k} \leq x \leq 3 - \frac{1}{k}\}, k=1, 2, 3, \dots$

(b) $C_k = \{(x, y): \frac{1}{k} \leq x^2 + y^2 \leq 4 - \frac{1}{k}\}, k=1, 2, 3, \dots$

Solution (a)

For every $k=1, 2, \dots$ $C_k \subset C_{k+1}$ (increasing) because

$$\frac{1}{k+1} < \frac{1}{k} \text{ \& } 3 - \frac{1}{k} < 3 - \frac{1}{k+1} \Rightarrow \left[\frac{1}{k}, 3 - \frac{1}{k}\right] \subset \left[\frac{1}{k+1}, 3 - \frac{1}{k+1}\right]$$

by 1.2.16

$$\lim_{k \rightarrow \infty} C_k = \left(\lim_{k \rightarrow \infty} \frac{1}{k}, \lim_{k \rightarrow \infty} 3 - \frac{1}{k} \right) = (0, 3) = \bigcup_{k=1}^{\infty} C_k$$

Note: 0 and 3 are not included

Solution (b)

For every $k=1, 2, 3, \dots$ $C_k \subset C_{k+1}$ (increasing)

$$\frac{1}{k+1} < \frac{1}{k} \text{ and } 4 - \frac{1}{k} < 4 - \frac{1}{k+1} \Rightarrow C_k \subset C_{k+1}$$

$$\lim_{k \rightarrow \infty} C_k = \{(x, y): 0 < x^2 + y^2 < 4\} = \bigcup_{k=1}^{\infty} C_k$$

by 1.2.16

Example 1.2.3

Suppose $C = \{1, 2, 3, \dots\}$. If $A_n = \{1, 3, \dots, 2n-1\}$
and $B_n = \{n, n+1, \dots\}$ for $n=1, 2, 3, \dots$ then

Since $A_n \subset A_{n+1}$ for $n=1, 2, 3, \dots$

A_n is an increasing sequence. Hence

$$\bigcup_{n=1}^{\infty} A_n = \lim_{n \rightarrow \infty} A_n = \{1, 3, 5, \dots\}$$

and

$$\bigcap_{n=1}^{\infty} A_n = A_1 = \{1\}$$

$$A_1 = \{1\}$$

$$A_2 = \{1, 3\}$$

$$A_3 = \{1, 3, 5\}$$

$\vdots$

Also, B_n is a decreasing sequence as $B_{n+1} \subset B_n$.

$$\text{Hence } \bigcap_{n=1}^{\infty} B_n = \lim_{n \rightarrow \infty} B_n = \emptyset$$

$$B_1 = \{1, 2, 3, \dots\}$$

$$B_2 = \{2, 3, \dots\}$$

$$B_3 = \{3, 4, \dots\}$$

$\vdots$

and,

$$\bigcup_{n=1}^{\infty} B_n = B_1 = C = \{1, 2, 3, \dots\}$$

Example 1.2.4

Suppose C is the interval of real numbers $(0, 5)$

Suppose $c_n = (1 - \frac{1}{n}, 2 + \frac{1}{n})$ and $D_n = (\frac{1}{n}, 3 - \frac{1}{n})$ for $n=1, 2, \dots$

Then

C_n is a decreasing sequence since $C_{n+1} \subset C_n$

Hence $\bigcap_{n=1}^{\infty} C_n = \lim_{n \rightarrow \infty} C_n = [1, 2]$

and

$$\bigcup_{n=1}^{\infty} C_n = C_1 = (0, 3)$$

$$C_1 = (0, 3)$$

$$C_2 = (\frac{1}{2}, \frac{5}{2})$$

$$C_3 = (\frac{2}{3}, \frac{7}{3})$$

$\vdots$

Also, D_n is an increasing sequence as $D_n \subset D_{n+1}$

Hence $\bigcup_{n=1}^{\infty} D_n = \lim_{n \rightarrow \infty} D_n = (0, 3)$

and

$$\bigcap_{n=1}^{\infty} D_n = D_1 = (1, 2)$$

$$D_1 = (1, 2)$$

$$D_2 = (\frac{1}{2}, \frac{5}{2})$$

$$D_3 = (\frac{1}{3}, \frac{8}{3})$$

$\vdots$